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The Infrared Spectrum of χ Cygni near the Maximum Light

Hideo MAEHARA

Department of Astronomy, University of Tokyo, Tokyo

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Abstract

An analysis of χ Cygni was made in the infrared region of the spectrum observed near the 1964 light-maximum. The main results obtained are as follows: (1) Each atomic absorption line has two components, and the excitation temperatures derived from the long- and short-wavelength components are about 1900 ± 300°K and 3100 ± 800°K, respectively. (2) CN molecular lines are found at the wavelengths corresponding to the short-wavelength components of atomic lines. (3) Metallic oxide bands (TiO, ZrO, and VO) have radial velocity related to the long-wavelength components of atomic lines. (4) Emission lines of the Hα and CaII infrared triplet have the radial velocity such that the wavelength is nearly equal to the short-wavelength components of atomic lines.

From these results, we suggest a scheme of the atmosphere of χ Cygni in which the disturbance due to shock waves plays an important role.

1. *Introduction*

χ Cygni (S7, 1e:—S10, 1e:, $m_V=3.3-14.2$, $P=406.95$ days) is one of the most famous long-period variable stars. MERRILL (1947) found from the spectrum of this star that iron lines at excitation potential 1.0 eV are displaced toward longer wave- lengths by 2 km/sec with respect to the ultimate lines. FUJITA (1951) showed for atomic absorption lines that great velocities of recession correspond to higher excitation potentials. In the present paper, we will show that radial velocities from absorption lines of TiI and FeI decrease with decreasing excitation potentials (low excitation potential lines are shifted toward short-wavelengths with respect to high excitation potential lines), and that our results are consistent with those obtained by MERRILL and FUJITA.

Concerning the problem of the chemical compositions of χ Cyngi, FUJITA (1952) determined the abundances of some atoms and molecules by using the curve-of-growth method. SPINRAD and VARDYA (1966) discussed H_2O and CN molecules, the characteristic features of M- and C-type spectra, respectively, and showed that this star has an intermediate chemical composition between types M and S as well as a high nitrogen abundance. We have found rotational lines of CN molecule in the spectrum of this star.

Concerning the variation with phase, MERRILL (1947, 1953) and FUJITA (1954) investigated the behavior of both absorption and emission lines mainly in the declining phase of the light-curve, and found some interesting results. Among these, the following conclusions seem to be much concerned with the light variation and the atmospheric structure.

(1) In the post-maximum phase, the radial velocities from the emission lines of FeI increase slightly with excitation potentials and phase, and most of

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them grew stronger with phase.

(2) Numerous emission lines of FeII become outstanding with advancing phase.

(3) Forbidden lines of FeII appear about 100 days after the maximum and increase their strength rapidly during the next two months.

(4) The absorption lines of CaI λ4227 and TiO λ5167 band are weaker in the post-maximum phase than at the light-maximum.

In this study, we will discuss double-line phenomena of χ Cygni, which is closely related to the atmospheric structure of the star.

2. *Observation and Measurements*

The spectrograms used in this study were obtained with the coudé spectrograph attached to the 188 cm telescope of the Okayama Astrophysical Observatory. Data of the spectrograms are given in Table 1. Figure 1 shows the

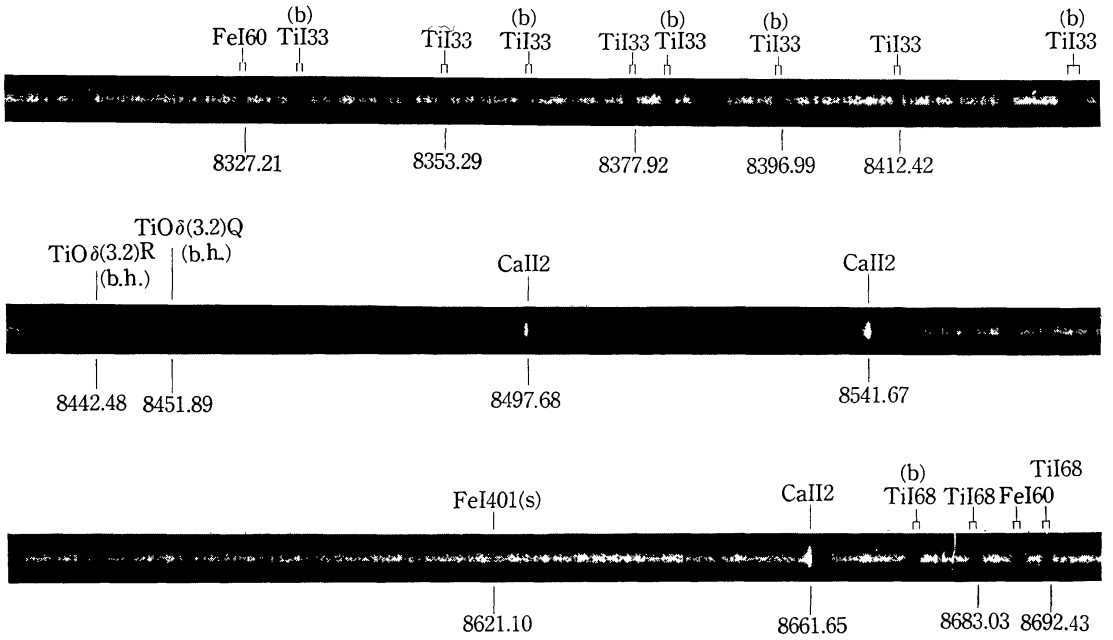

FIG. 1. A portion of the spetrogram C10-860. Strong atomic lines and bands are indicated. (b): Blended with other lines. (s): The short wavelength component.

reproduction of plate CIO-860. As the spectrum of this star shows considerable variation from cycle to cycle as well as with phase of the light variation (KEENAN 1966), the spectrograms obtained at different phases and cycles should not be

TABLE 1. Data of Spectrograms.

Plate No.	Date	Phase	Emulsion	Dispersion	Spectral Region
C10-860	1964 Mar. 5	+10 days	IN*	8 A/mm	$\lambda\lambda$ 7900-8750
C10-863	1964 Mar. 7	+12	IN*	8	7200-8100
C10-865	1964 Mar. 8	+13	103a-F	8	6200-6900

*: Ammonia sensitized.

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treated identically. The spectrograms in Table 1 were obtained on four successive nights, and therefore, they should be considered to represent a spectrum at one particular phase (about +10 days after the estimated 1964 maximum).

A microphotometric tracing of C10-860 with low magnification is shown in Figure 2, where the ordinate is the density-scale and the abscissa is the wavelength in Angstroms. As seen in this figure, there are many spectral regions where band absorptions are intense, and the location of the continuum is rather ambiguous.

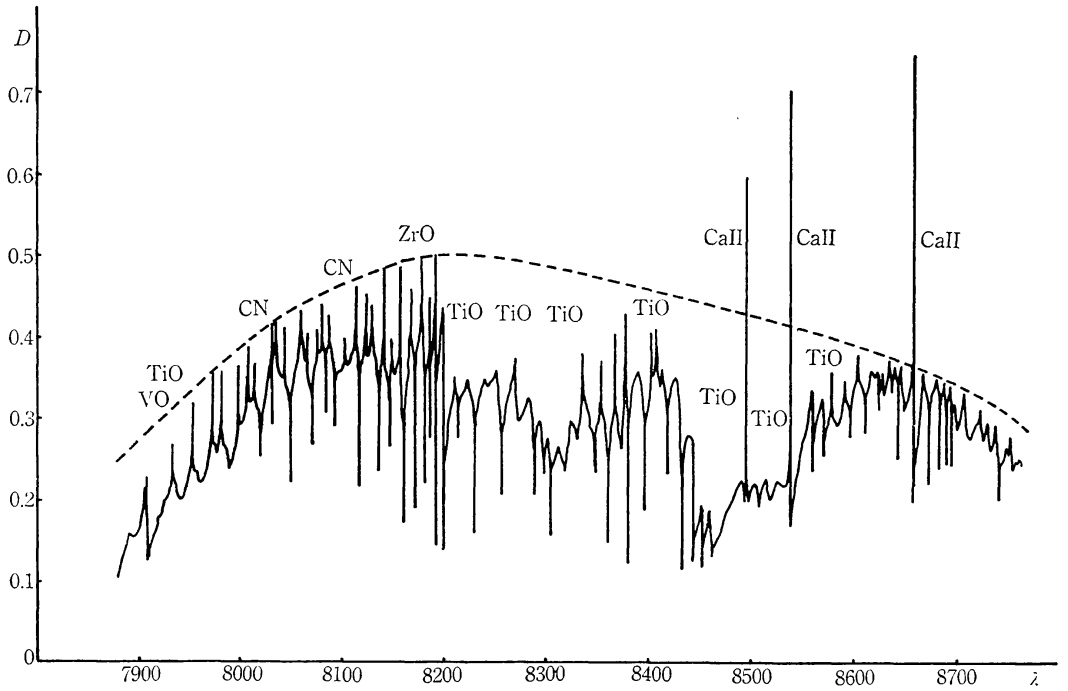

FIG. 2. The microphotometric tracing with low magnification.

The wavelengths of the spectral lines were measured by a normal travelling micrometer and their identifications were made with the aid of the following references;

(1) atomic lines: MOORE (1945), CORLISS and BOZMAN (1962), DAVIS (1947), and HARRISON (1939).

(2) CN molecules DAVIS and PHILLIPS (1963), and WYLLER (1966).

(3) metallic oxides: DAVIS (1947), and GATTERER, JUNKES and SALPETER (1957).

(4) telluric lines: BABCOCK and MOORE (1947).

3. *Atomic Absorption Lines*

In the course of the line identification, it was found that strong atomic absorption lines are double, especially for wavelengths longer than λ 8000. Figure 3 shows a portion of the intensity tracing with high magnification. The double components of the absorption lines are clearly seen in this figure and Figure 1.

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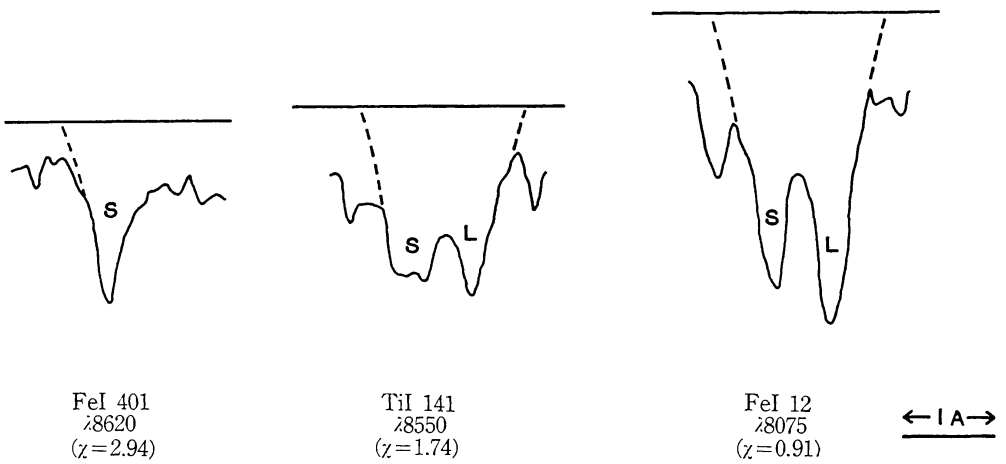

FIG. 3. A portion of the intensity tracing with high magnification. S means the *s*-component of an atomic line, and L means the *l*-component.

TABLE 2. Radial Velocities from TiI Lines.

λ_*	λ_{lab}	Component	Mult. No.	χ (eV)	V (km/sec)
7908.96	7909.34	*s*	308	3.30	−14
7938.06	7938.53	*s*	151	1.87	−18
7938.64	7938.53	*l*	151	1.87	+ 4
7943.38	7943.93	*s*	308	3.28	−21
7948.51	7949.17	*s*	125	1.50	−25
7949.15	7949.17	*l*	125	1.50	− 1
8024.93	8024.84	*l*	151	1.87	+ 3
8067.59	8068.24	*s*	151	1.87	−24
8353.29	8353.15	*l*	33	0.81	+ 5
8364.32	8364.24	*l*	33	0.83	+ 3
8377.31	8377.90	*s*	33	0.82	−21
8377.92	8377.90	*l*	33	0.82	+ 1
8402.60	8402.54	*l*	224	2.24	+ 2
8411.74	8412.36	*s*	33	0.81	−22
8412.42	8412.36	*l*	33	0.81	+ 2
8423.84	8424.41	*s*	182	2.09	−20
8426.59	8426.50	*l*	33	0.82	+ 3
8435.69	8435.68	*l*	33	0.83	0
8438.36	8438.93	*s*	224	2.25	−20
8538.66	8539.36	*s*	209	2.23	−25
8547.62	8548.07	*s*	150	1.87	−16
8550.07	8550.54	*s*	141	1.74	−16
8550.72	8550.54	*l*	141	1.74	+ 6
8612.18	8612.91	*s*	141	1.73	−25
8612.96	8612.91	*l*	141	1.73	+ 2
8674.82	8675.38	*s*	68	1.06	−19
8675.47	8675.38	*l*	68	1.06	+ 3
8682.35	8682.99	*s*	68	1.05	−22
8683.03	8682.99	*l*	68	1.05	+ 1
8691.63	8692.34	*s*	68	1.04	−25
8692.43	8692.34	*l*	68	1.04	+ 3

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TABLE 3. Radial Velocities from FeI Lines.

λ_*	λ_{lab}	Component	Mult. No.	χ (eV)	V (km/sec)
7912.20	7912.87	*s*	12	0.86	−25
7912.97	7912.87	*l*	12	0.86	+ 4
7940.67	7941.09	*s*	623	3.26	−16
8045.56	8046.07	*s*	1136	4.40	−19
8046.99	8047.60	*s*	12	0.86	−23
8047.73	8047.60	*l*	12	0.86	+ 5
8071.71	8072.16	*s*	108p	2.41	−17
8074.52	8075.13	*s*	12	0.91	−23
8075.25	8075.13	*l*	12	0.91	+ 4
8238.62	8239.13	*s*	108	2.41	−19
8247.73	8248.15	*s*	1136	4.35	−15
8292.95	8293.53	*s*	623	3.29	−21
8326.45	8327.06	*s*	60	2.19	−22
8327.21	8327.06	*l*	60	2.19	+ 5
8331.44	8331.94	*s*	1153	4.37	−18
8341.78	8342.21	*s*	401p	2.94	−15
8387.77	8387.78	*l*	60	2.17	0
8571.40	8571.84	*s*	1272p	4.99	−15
8598.30	8598.79	*s*	1153	4.37	−17
8606.53	8607.08	*s*	1272p	4.99	−19
8611.17	8611.81	*s*	339	2.83	−22
8621.10	8621.61	*s*	401	2.94	−18
8651.99	8652.50	*s*	1050p	4.14	−18
8674.08	8674.75	*s*	339	2.82	−23
8688.00	8688.63	*s*	60	2.17	−22
8688.66	8688.63	*l*	60	2.17	+ 1

(1) *Radial Velocities of the Double Lines*

Tables 2 and 3 give the radial velocity derived from both of the component lines. Each component of the double absorption lines gives a fairly constant radial velocity. Figures 4 and 5 show a relation between the radial velocity and the excitation potential; the abscissae give the excitation potential of the level from which the line originates, and the ordinates the heliocentric radial velocity of each component line. The upper group of points in Figures 4 and 5 corresponds to the long-wavelength component of the lines, and the lower group to the short-wavelength component (hereafter, they will be called the *l*-component and the *s*-component, respectively).

The radial velocity of the *s*-component lines increases with increasing excitation potential, χ, of the lower level. For the *s*-components of FeI (Figure 5), the following relation is obtained,

$$V=(1.6\pm0.3)\chi-(24.6\pm1.0)\ ,$$

where χ is in electron-volts and V is the radial velocity in km/sec. As is shown in Figure 4, a similar relation is also obtained for the *s*-components of TiI lines.

Though the *l*-component lines of low excitation potential are quite strong,

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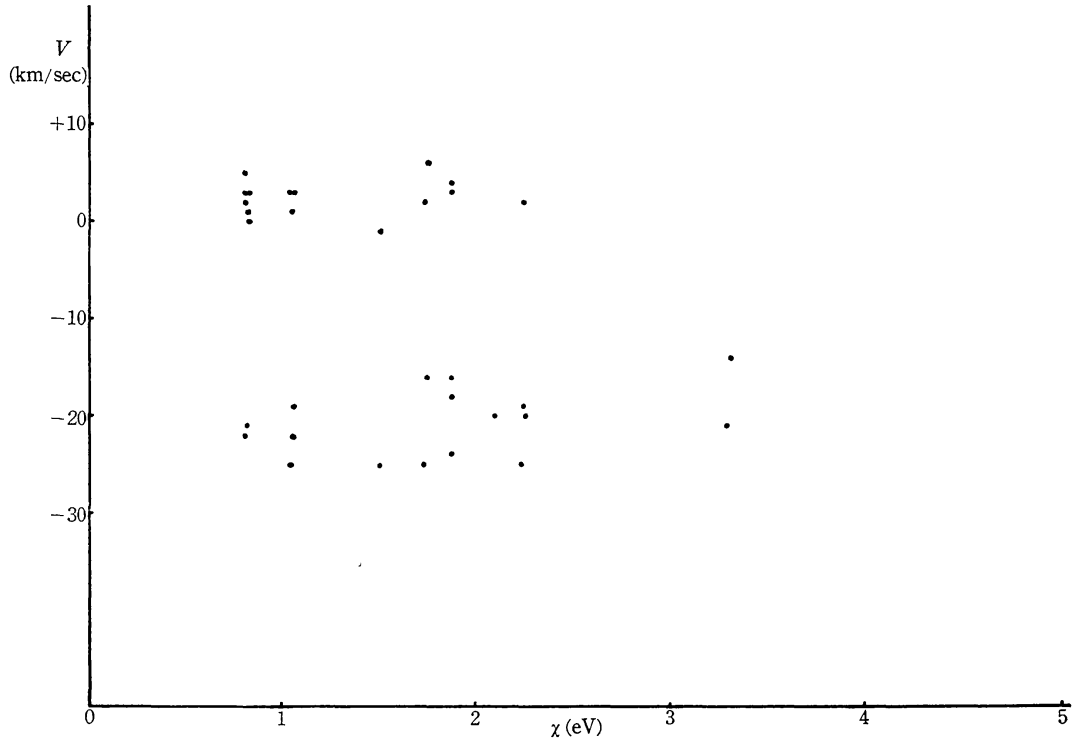

FIG. 4. The heliocentric radial velocity from TiI lines versus the excitation potential of the lower level (eV).

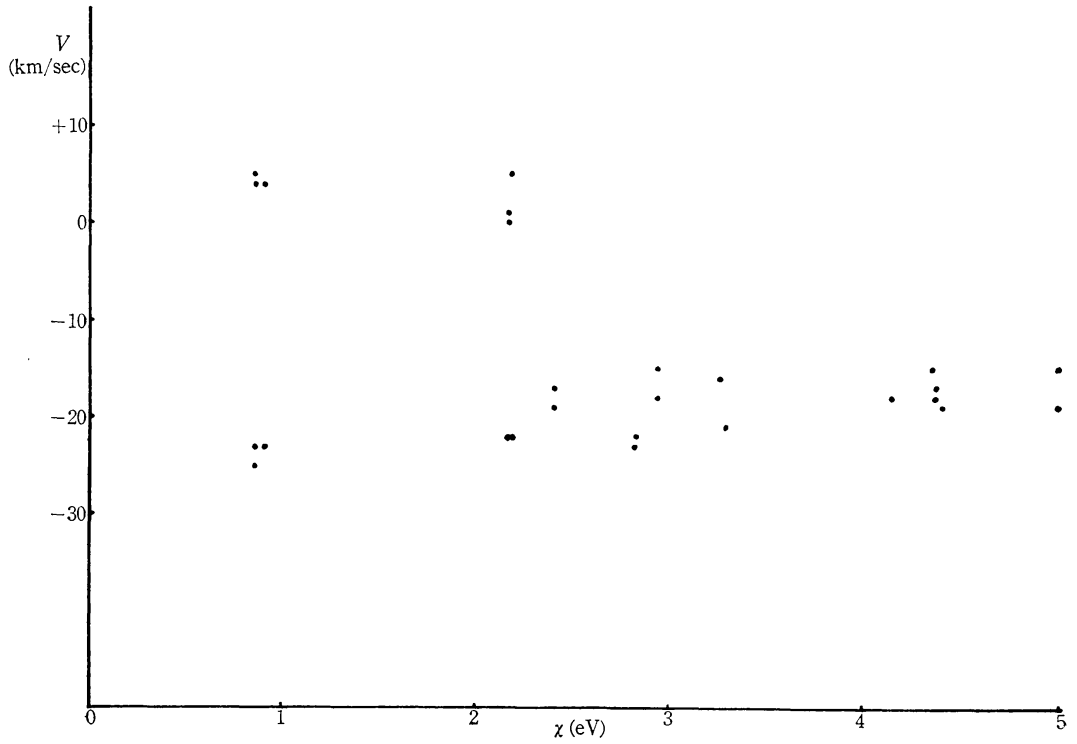

FIG. 5. The heliocentric radial velocity from FeI lines versus the excitation potential of the lower level (eV).

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few lines of high excitation potentials could be seen because their intensities decrease rapidly with increasing excitation potentials. From this situation, no precise relation could be derived between the radial velocity and the excitation potential for the *l*-components.

It is kown for long-period variables that the radial velocity of absorption lines depends on the excitation potential. In particular, extensive studies on the radial velocity problem were carried out for the following stars: *o* Ceti (JOY 1954), R Hydrae (MERRILL 1946a), R Leonis (MERRILL 1946b), R Andromedae (MERRILL and GREENSTEIN 1956), and χ Cygni (MERRILL 1947; FUJITA 1951). On the other hand, double absorption lines are found in *o* Ceti (ADAMES 1941) and R Andromedae (MERRILL and GREENSTEIN 1958; SPINRAD 1965; TSUJI 1966). MERRILL (1960) thought that the χ-dependence of the radial velocity is a minor manifestation of the double lines. However, according to the present results, we conclude these two phenomena are independent of each other.

The χ-dependence of the radial velocity of the *s*-components as well as the double lines indicate the stratification in the atmosphere.

(2) *Intensities of the Double Lines*

Intensities of the double lines also show an interesting variation with the excitation potential of the lower level, as is seen in Figure 3. The *s*-component lines with low excitation potentials are weaker (the depths of the strongest lines $R\approx 0.5$), and the *s*-components with high excitation potentials are rather stronger than the spectral lines in such stars as XY Lyrae (M 5) and 57 Pegasi (M 4 S). On the other hand, the *l*-component lines with low excitation potentials are quite deep ($R\approx 1.0$), and those with excitation potentials of more than 2 eV are too weak to be seen. It is suggested from these results that the *s*-components are formed in a high temperature layer, and the *l*-components are in a low temperature layer.

4. *Temperature Determination by the Curve-of-Growth Method*

In order to determine the excitation temperature, we constructed the curve of growth for each component of TiI, FeI, and other elements. The lines used are in the regions $\lambda\lambda$ 7900–8100 and around λ 8600 : in both regions, band absorption is weak and the location of the continuum is determined fairly well (see Figure 2). Since many lines are blended with the other components and other atomic and molecular lines, we measured the central depths of the lines instead of the equivalent widths.

Pseudo-curves-of-growth constructed for TiI lines are shown in Figures 6 and 7. We plotted $\log R$ (R is the central depth of a line) versus $\log X_f$, where

$$\log X_f = \log gf\lambda - \chi\theta \ .$$

The notation is that of general use. The gf-values were referred to CORLISS and WARNER (1964) for FeI lines, and to CORLISS and BOZMAN (1962) for other atomic lines. Thus, we can calculate $\log X_f$ by specifying the reciprocal temperature θ. The excitation temperature is determined by θ at which all dots in the curve of growth show the minimum scatter. In actual procedure, however, the number of available lines was small and the position of the continuum was somewhat uncertain; therefore, only a rough value could be obtained for the

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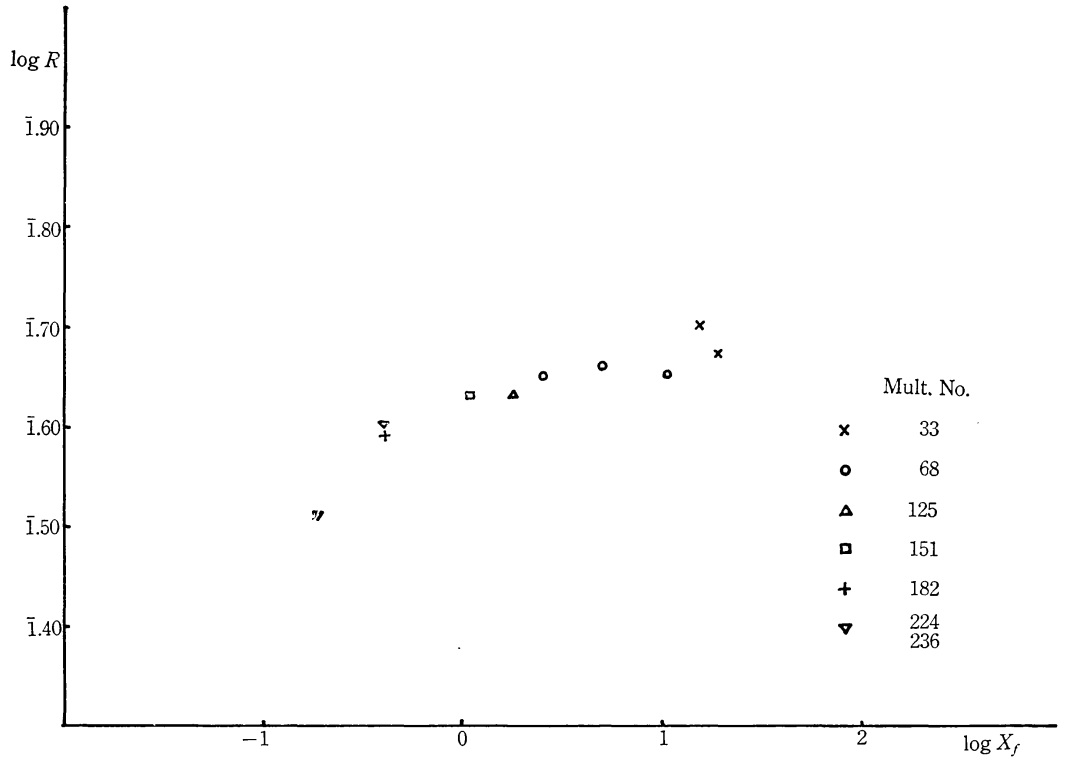

FIG. 6. Curve of growth for the *s*-component lines of TiI. R is the central depth of a line, and $\log X_f = \log gf\lambda - \chi\theta$.

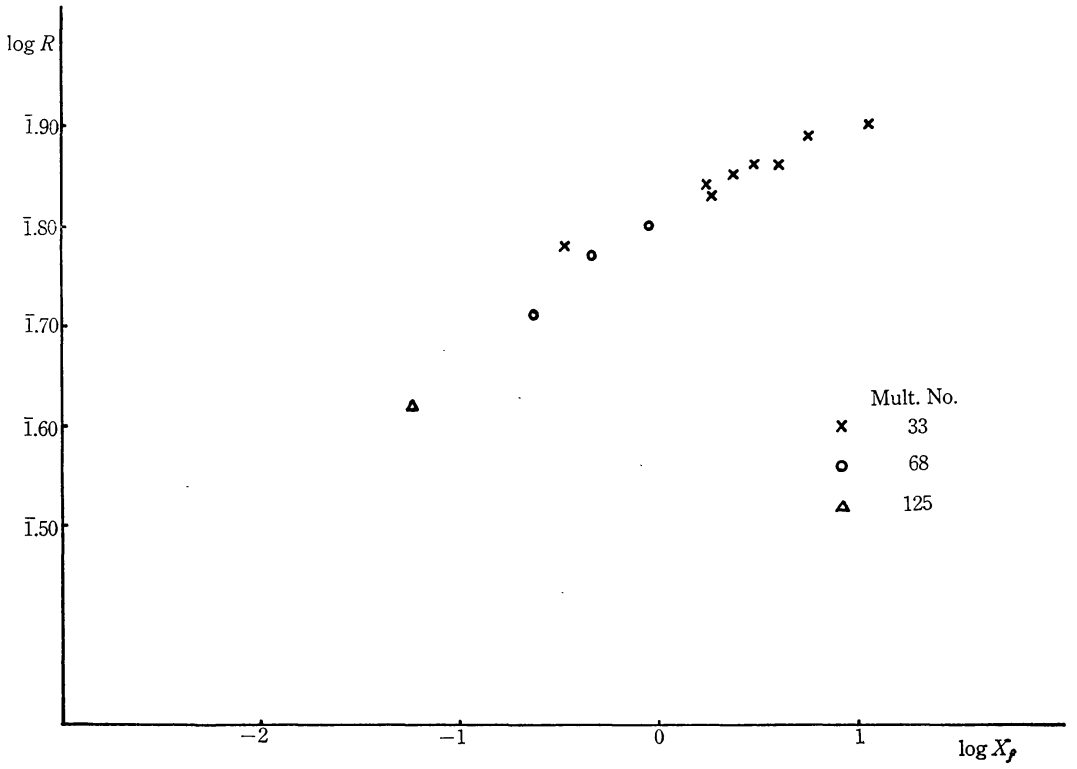

FIG. 7. Curve of growth for the *l*-component lines of TiI. The caption is the same as that of Figure 6.

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excitation temperature from FeI and TiI lines:

$\theta=1.6\pm0.4$ ($T=3100\pm800$°K) for the s-components

and $\theta=2.6\pm0.4$ ($T=1900\pm300$°K) for the l-components.

5. *Molecular Absorption*

(1) *Rotational Lines of Cyanogen*

According to DAVIS and PHILLIPS (1963), we identified rotational lines of CN molecule in the region $\lambda\lambda$ 7900–8200 where absorption of metallic oxides is weak and CN lines are relatively strong. Their radial velocity coincides with that of the s-components of atomic lines. Figure 8 shows a portion of the

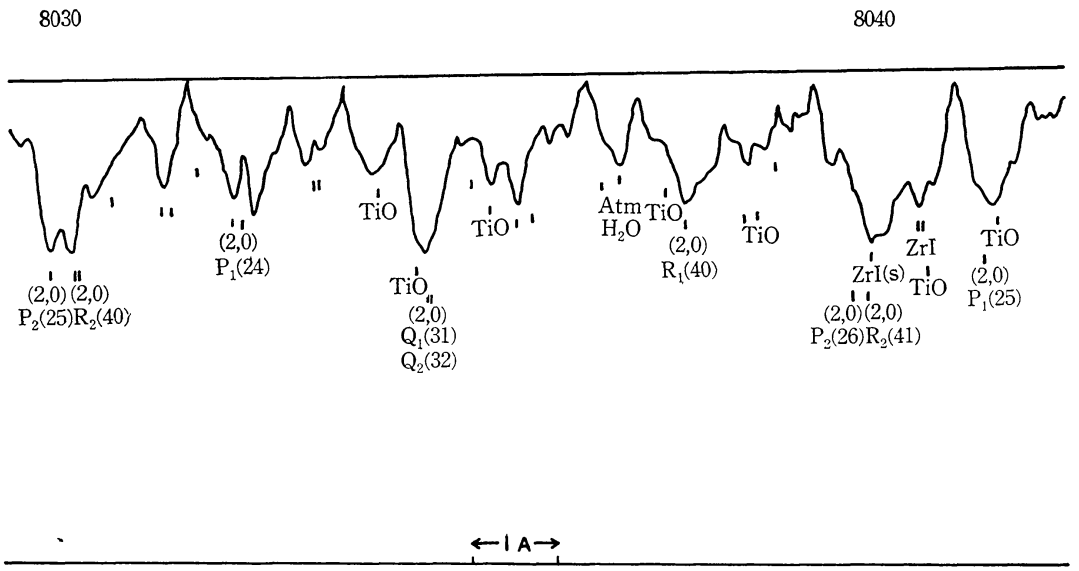

FIG. 8. A portion of the microphotometric tracing, showing CN lines. The dashes indicate weak CN lines.

microphotometric tracing. (CN lines as well as other conspicuous absorption features are marked. Weak CN lines are indicated by dashes in this figure.) Figures 9 and 10 show a chance-coincidence test for rotational lines of the R_2 branch of CN (3, 1) band. Line depths in the intensity tracing were measured (regardless of whether a line exists or not) at the wavelength at which a CN line was expected on both the wavelength systems derived from the s-components and the l-components of atomic lines. Figure 9 shows the plots of the depths at the wavelength on the system based on the s-components versus the rotational quantum number K. Figure 10 shows the same plots for the system based on the l-components. As is shown in these figures, the s-components of CN are present but the l-components are absent or very weak.

From the theory of dissociative equilibrium (TSUJI 1964), CN is most abundant at $\theta=1.4$ ($T=3600$°K) in normal giant stars of types M, S, and C, and less abundant both at higher and lower temperature. Thus, the results mentioned above seem to be consistent with the temperatures obtained in Section 4.

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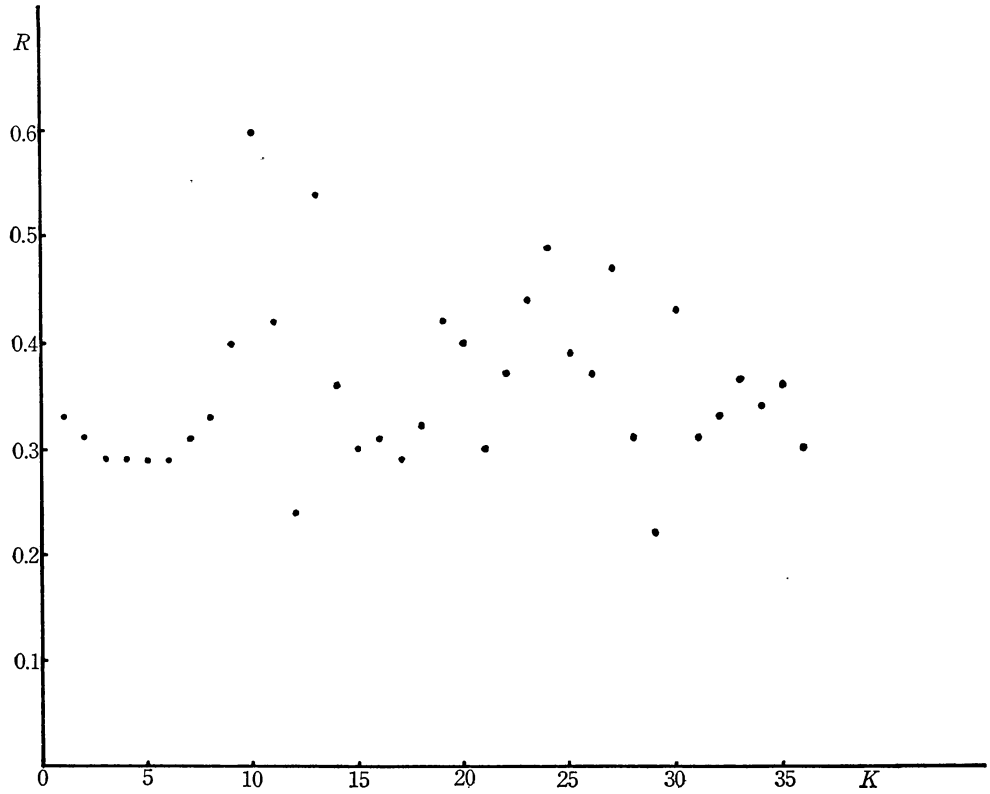

FIG. 9. A chance-coincidence test for the R_2 branch of CN (3, 1) band. The dots show the depths in the intensity tracing at the wavelength on the system based on the *s*-components versus the rotational quantum number *K*.

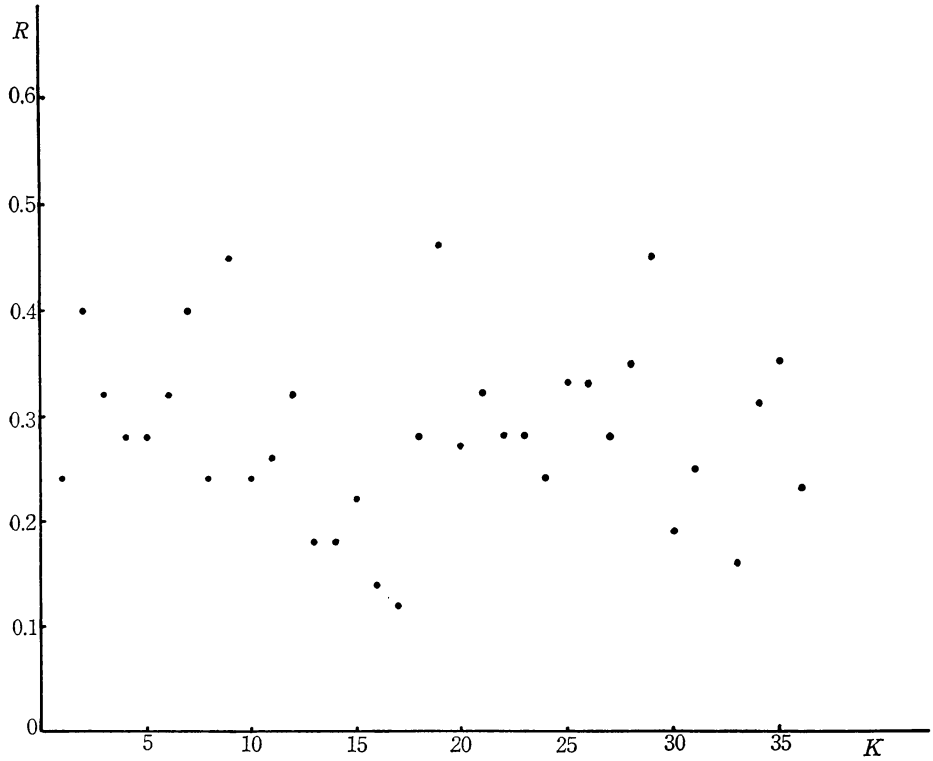

FIG. 10. A chance-coincidence test for the R_2 branch of CN (3, 1) band. The dots show the depths at the wavelength on the *l*-components versus the rotational quantum number *K*.

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(2) *Bands of Metallic Oxides*

Within the spectral region covered in this study, TiO bands are dominant in the regions λλ 6600–6900, λλ 7600–8000, and λλ 8200–8500. Moreover, strong ZrO bands exist in the region shorter than λ 6600, and weak VO bands are found around λ 7900. YO, ScO, AlO etc. have been found in the spectrum of χ Cygni, but they are not seen in the region covered in this investigation.

TABLE 4. Radial Velocities from TiO Band-Heads (γ-System) in the Region λλ 7900–8700.

Band	Branch	λ_*	λ_{lab}	V (km/sec)
(3,4)	R_1	7907.50	7907.5	0
(0,2)	R_3	8205.91	8205.8	+ 4
(0,2)	R_2	8250.77	8250.6	+ 6
(1,3)	R_3	8289.13	8289.0	+ 5
(1,3)	R_2	8334.60	8334.5	+ 4
(1,3)	R_1	8386.76	8386.5	+ 9
(2,4)	R_2	8419.72	8419.5	+ 8
(3,2)	R	8442.48	8442.3	+ 6
(3,2)	Q	8451.89	8451.7	+ 7
(2,4)	R_1	8471.83	8471.6	+ 8
(3,5)	R_2	8505.58	8505.5	+ 3
(3,5)	R_1	8558.65	8558.4	+ 9
(4,6)	R_1	8645.70	8645.5	+ 7

As is shown in Table 4, radial velocities derived from band-heads coincide with those of the *l*-components of atomic lines. Band-heads where the radial velocity corresponds to that of the *s*-components are not seen in the spectrum. It is known from discussion of dissociative equilibrium that metallic oxides usually tend to be more abundant at low temperature (this tendency, however, depends on the molecule considered, H : C : N : O abundance ratio, and the pressure of the atmosphere) and therefore, the layer of the *l*-components of atomic lines has lower temperature than that of the *s*-components.

6. *Emission Lines*

Hα and CaII infrared triplet lines were observed in emission. Since all of them are multilated by strong TiO absorption, their radial velocities, half-widths and intensities may be affected seriously by the absorption effect. The results of the measurements are given in Table 5, and their spectral features are represented in Figures 11 and 12.

TABLE 5. Emission Lines.

λ_*	Identification	V (km/sec)	Half-Width	Int. (Cont.=1)
6562.24	H α	−27	1.0 A	2
8497.68	CaII 8494	−12	0.5	1.9
8451.67	CaII 8542	−15	0.5	2.7
8661.65	CaII 8662	−17	0.4	3.6

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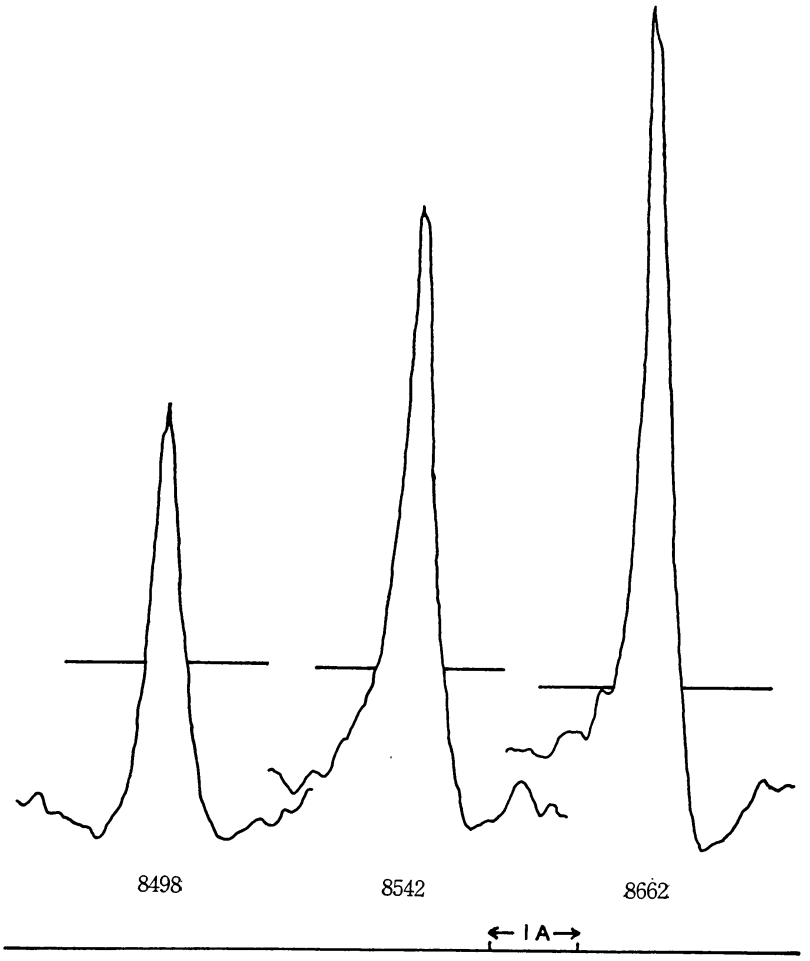

FIG. 11. The intensity tracing of CaII infrared emission lines.

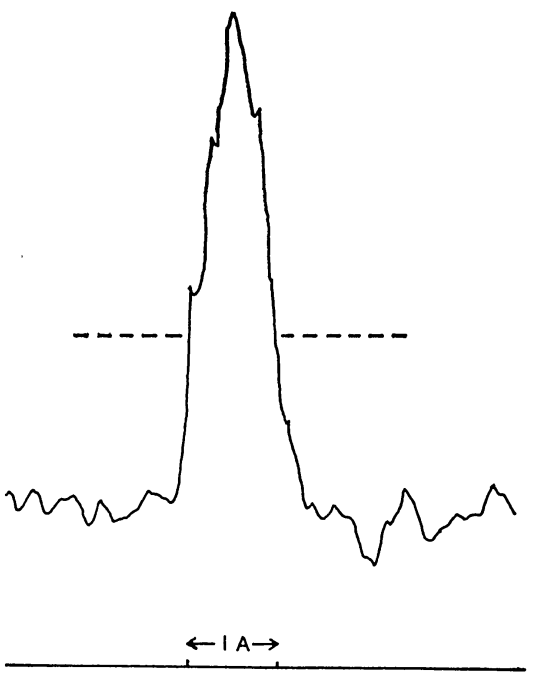

FIG. 12. The intensity tracing of Hα emission line.

The intensities of CaII emission lines differ markedly from the laboratory values, but this may not be real because the absorption bands of TiO have a serious effect on them. The effect of TiO bands is strongest at λ 8498, next strongest at λ 8542, and not so strong at λ 8662. As is shown in Figure 11, the absorption lines due to CaII itself are found at wavelengths corresponding to the *l*-component lines. Hα emission line is much mutilated by strong TiO and ZrO bands. Absorption is so strong around λ 6600 that we cannot determine the location of the continuum.

Radial velocities of these emission lines are nearly equal to (but somewhat greater than) those of the *s*-components of atomic lines. In

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general, it is well established, in the spectra of the long-period variables in the visual to photographic region, that the radial velocity of the emission lines is smaller than that of the absorption lines. These absorption lines seem to correspond to the *l*-components of the double absorption lines in this study.

7. *Discussion*

We may expect the following cases in which the double absorption lines can be observed:

(1) a binary system (double-lined binary),
(2) an apparent separation due to emission (self-reversal),
(3) a single star which has two layers of different radial velocities.

The light-curves of χ Cygni cannot be interpreted as a binary system, and there are no other evidences indicating that this star has a companion. Secondly, the intensity variation of atomic lines with the excitation potential, and the difference in the radial velocities of cyanogen and metallic oxides, as stated respectively in Sections 3 and 5, cannot be explained by self-reversal. Moreover, if we suppose that both components of atomic lines are formed in a single layer, their combined line widths seem to be too broad. Thus cases (1) and (2) are improbable.

KAMIJO (1962) showed from the theoretical approach that a progressive wave type pulsation may possibly exist in the long-period variables. DEUTSCH and MERRILL (1959) and GORBATSKII (1961) discussed the case in which the shock wave is responsible for the emission lines. Considering these factors, we conclude that the double absorption lines are formed in two layers of different radial velocities due to the shock wave.

The next problem is to locate two layers in the atmosphere. Hereafter, the "*l*-layer" means the layer where the *l*-components of atomic lines are formed, and the "*s*-layer" means the layer responsible for the *s*-components. The *s*-components of atomic lines and the CN lines are formed in the *s*-layer, and the *l*-components of atomic lines and the bands of TiO, ZrO, and VO are

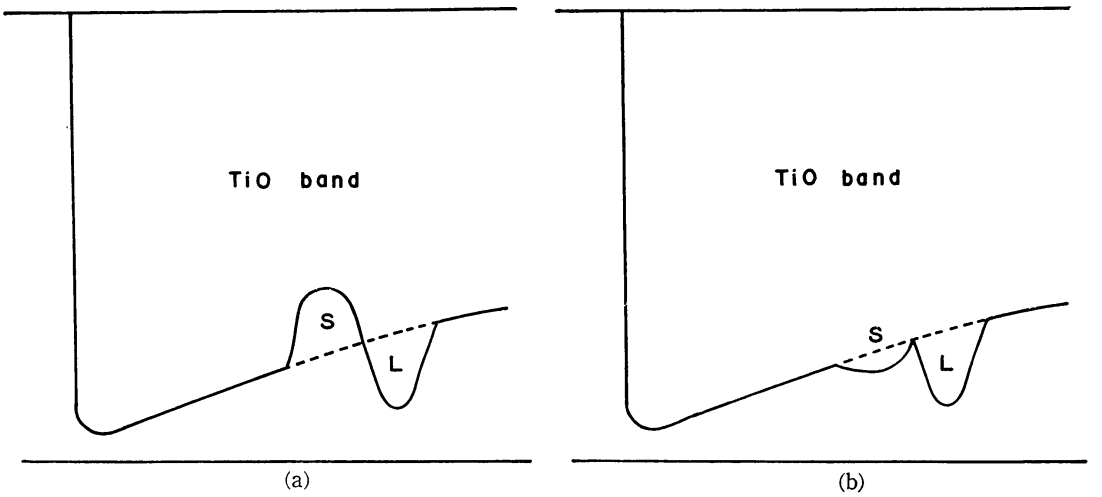

FIG. 13. An illustration made in order to distinguish the location of the *s*- and *l*-layer.
(a) Expected features in the case where the *s*-layer is outside the *l*-layer.
(b) Expected features in the case where the *s*-layer is inside the *l*-layer.

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found in the l-layer. The problem can be solved by considering the strong atomic lines in the spectral region where TiO bands are fairly strong (the depths are larger than 0.5) and their band profile forms a quasi-continuum. In these regions, the l-layer is optically thick due to the TiO bands, and the s-layer is also thick at the line center of the strong s-components considered.

If we suppose that the s-layer is outside the l-layer, the s-component lines are formed on the TiO continuum. Thus, they should appear in emission, because the s-layer is optically thick at the line center of the s-components and the depths of the TiO bands are larger than the central depths of the strongest s-components without band absorption ($R \approx 0.5$). The expected spectral feature is shown schematically in Figure 13(a). On the other hand, if the s-layer is inside the l-layer, the$s$-components are formed at first and then they would suffer from modulation by TiO bands. Therefore, they should be smeared out or appear in weak absorption, since the optical thickness of TiO is fairly large, as is shown in Figure 13(b).

In the actual spectrum of χ Cygni, all these s-component lines appear in weak absorption. Thus, we conclude that the s-layer is inside the l-layer. Moreover, we have obtained the spectrograms of χ Cygni in the 1967 cycle at phases -18 days, $+26$ days, and $+66$ days (referred to the estimated maximum). The data are given in Table 6 and the mean radial velocities of the double absorption lines and CaII emission lines are given in Table 7. Figure 14 shows the

TABLE 6. Data of Spectrograms.

Plate No.	Date	Phase	Emulsion	Dispersion	Spectral Region
C10-1664	1967 May 27	−18 days	IN*	8 A/mm	λλ 7900-8750
C10-1686	1967 July 10	+26	IN*	8	7900-8750
C10-1733	1967 Aug. 19	+66	IN*	8	7900-8750

TABLE 7. Mean Radial Velocities from the Double Absorption Lines (TiI and FeI) and CaII Emission Lines.

Plate No.	Radial Velocities (km/sec)		
	s-Components	l-Components	CaII Emission Lines
C10-1664	−20.8	+4.9	−12.6
C10-1686	−19.7	+5.3	−16.2
C10-1733	−18.0	+4.5	−15.3
(C10- 860	−20.0	+2.7	−14.7)

microphotometric tracing of the double lines. It is clearly seen in Table 7 and Figure 14 that the l-component lines are weakened with phase, and the radial velocity of the s-components increases while that of the l-components is constant (the separation of the l- and s-components decreases) with phase.

Now, we suggest the following scheme of the atmosphere of χ Cygni; a shock front exists at the boundary of the s- and l-layer, moving to the l-layer (or outwards). The l-layer is a cool, tenuous outer region at rest, and the s-layer is a relatively hot inner region behind the shock front moving outwards.

This scheme is suggested by the results obtained above:

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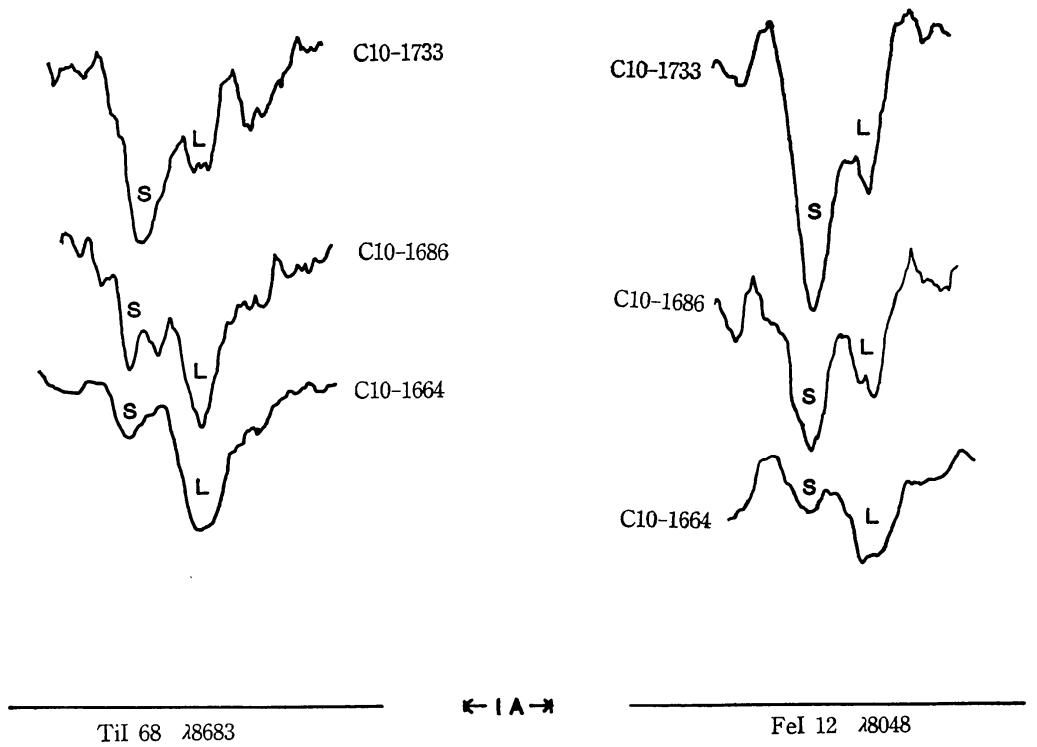

FIG. 14. The microphotometric tracing of the double lines. **S** and **L** mean the *s*- and *l*-component of an atomic line, respectively.

(1) The excitation temperature of the *s*-layer ($T=3100\pm800°$K) is higher than that of the *l*-layer ($T=1900\pm300°$K).

(2) The radial velocities of cyanogen and metallic oxides are nearly equal to those of the *s*- and *l*-layer, respectively.

(3) The radial velocity of the emission lines is nearly equal to that of the *s*-layer. (Since the layer where the emission lines are effectively formed has fairly high temperature, it must be a layer heated by the shock wave.)

(4) It is shown in the spectrograms of the 1967 cycle that the *l*-layer is at rest while the *s*-layer is decelerated, and the optical thickness of the *l*-layer decreases with phase.

In this paper, we mostly studied the spectrum near the maximum phase. Hence, the results obtained here should be re-examined and treated quantitatively in future investigations, and variation of the double lines and emission lines with phase should be studied in order to understand the whole phenomena taking place in this typical long-period variable star.

In conclusion, I wish to express my hearty thanks to Professor Y. Fujita for putting the spectrograms at my disposal and for his constant encouragement, and to Dr. Y. Yamashita for his valuable advice and discussion, and for reading the manuscript.

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